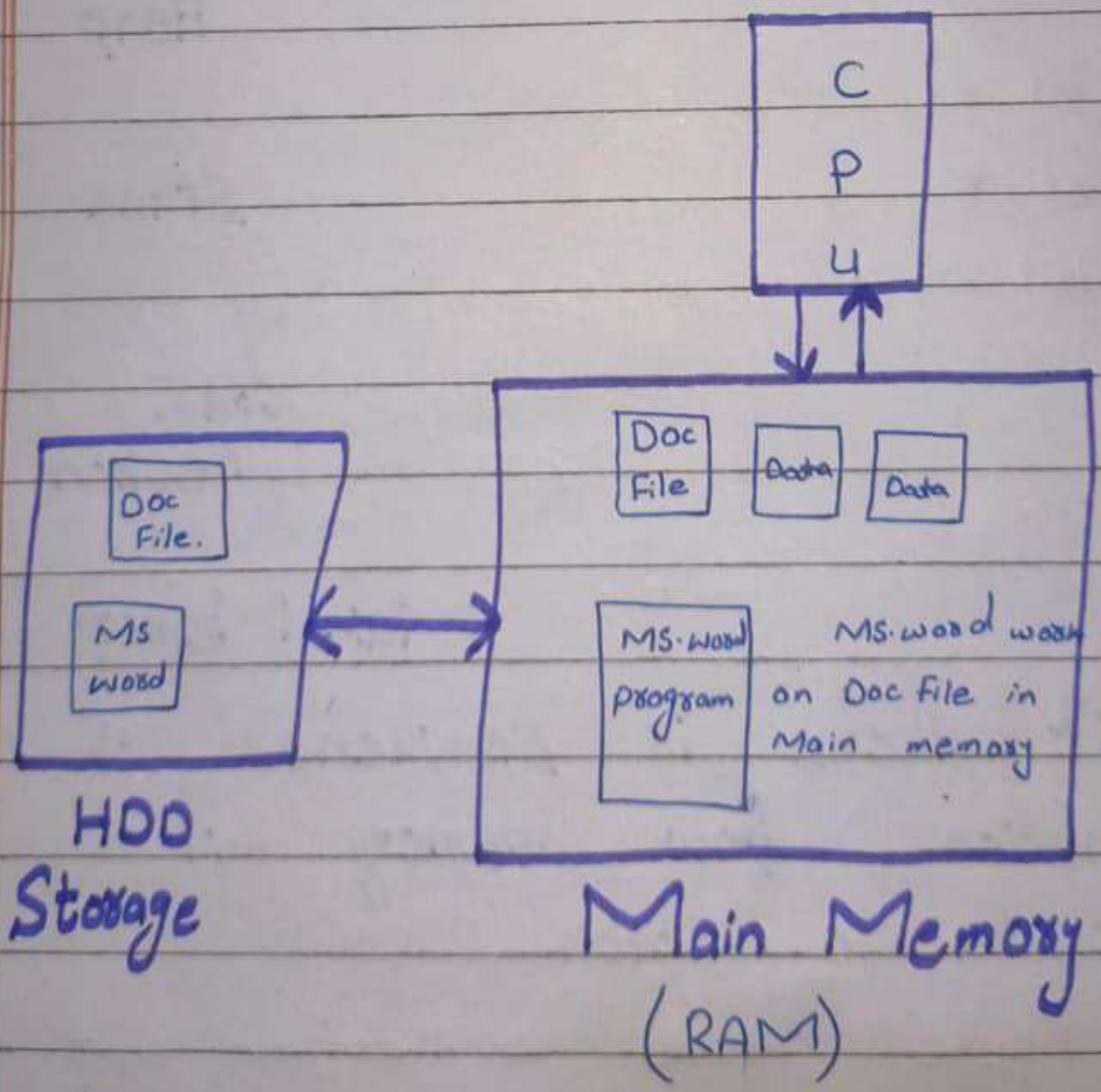
DATA STRUCTURE AND ALGORITHMS [DSA]

C/C++

Introduction of DSA

Def :- Arrangment of collection of Data item So the operation on data can be done inside the main memory during the Execution of a program.

Data Structure.



→ Arrangment of Data in Main Memory So it is utilize effeciently by Program is Called Data Structure.

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Static Vs Dynamic Memory Allocation

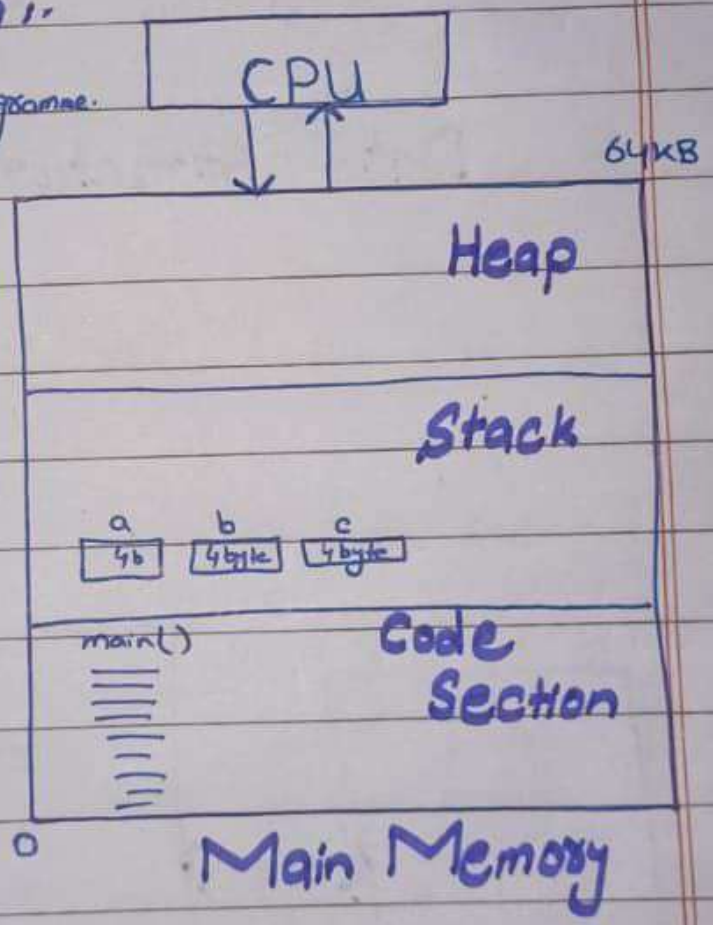
Static Memory Allocation:-

The size of the memory is static when it is done at Compile time.

Explaination:-

Lets make one programme.

```
int main()
{
  int a;
  int b;
  float c;
}
```



→ if there is sequence of function how memory will allocate in Stack.

Lets make a program that contain Some function ~~program~~.

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```
void fun1(int i){
    int a;
    }
```

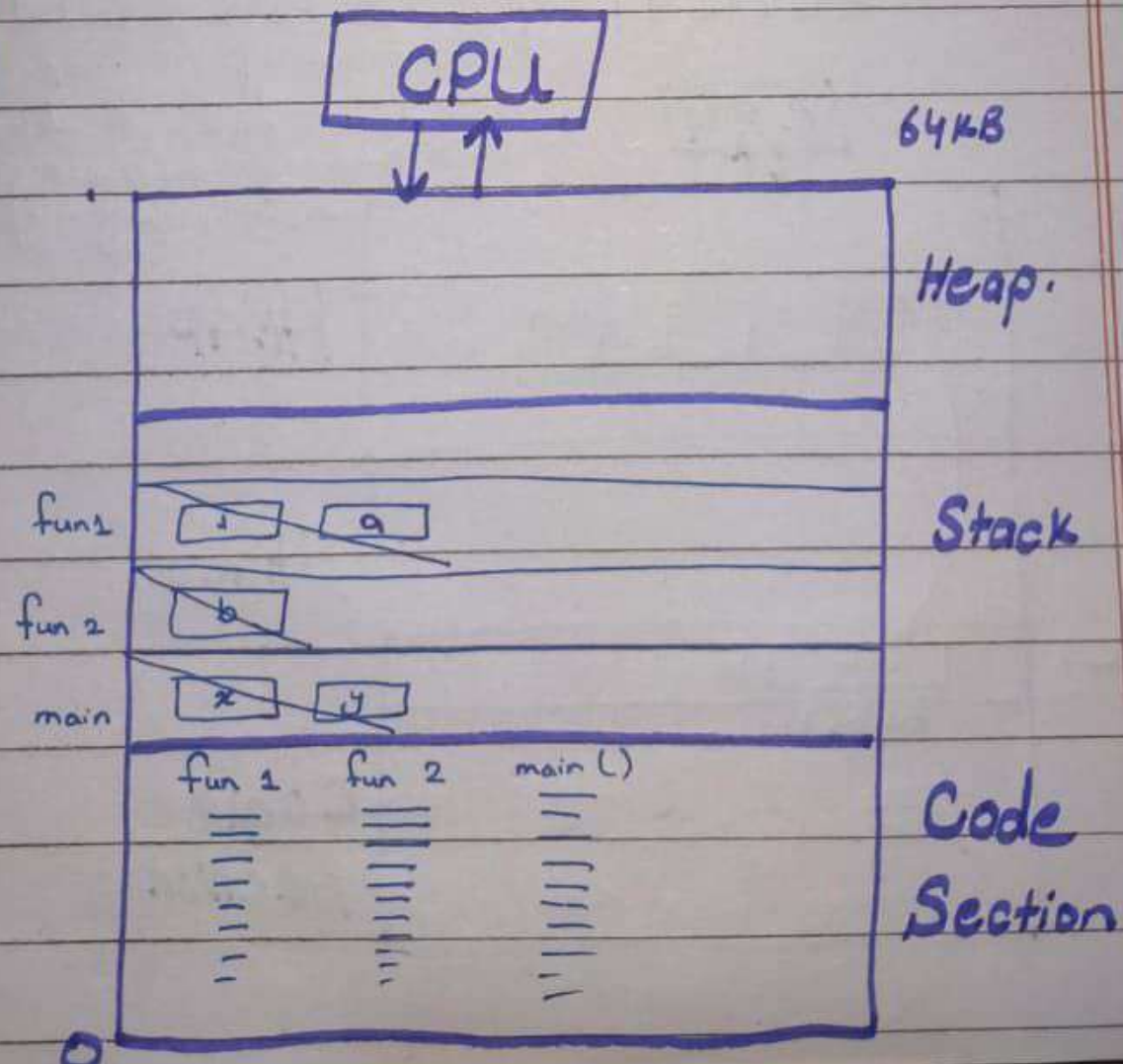
when fun1 will executed its memory will be deleted.

```
void fun2( ){
    int b;
    fun1(b);
}
```

Same for fun2

```
void main(){
    int x, y;
    fun2();
    cout << " OSA" << endl.
return 0;
```

Same for main fun.



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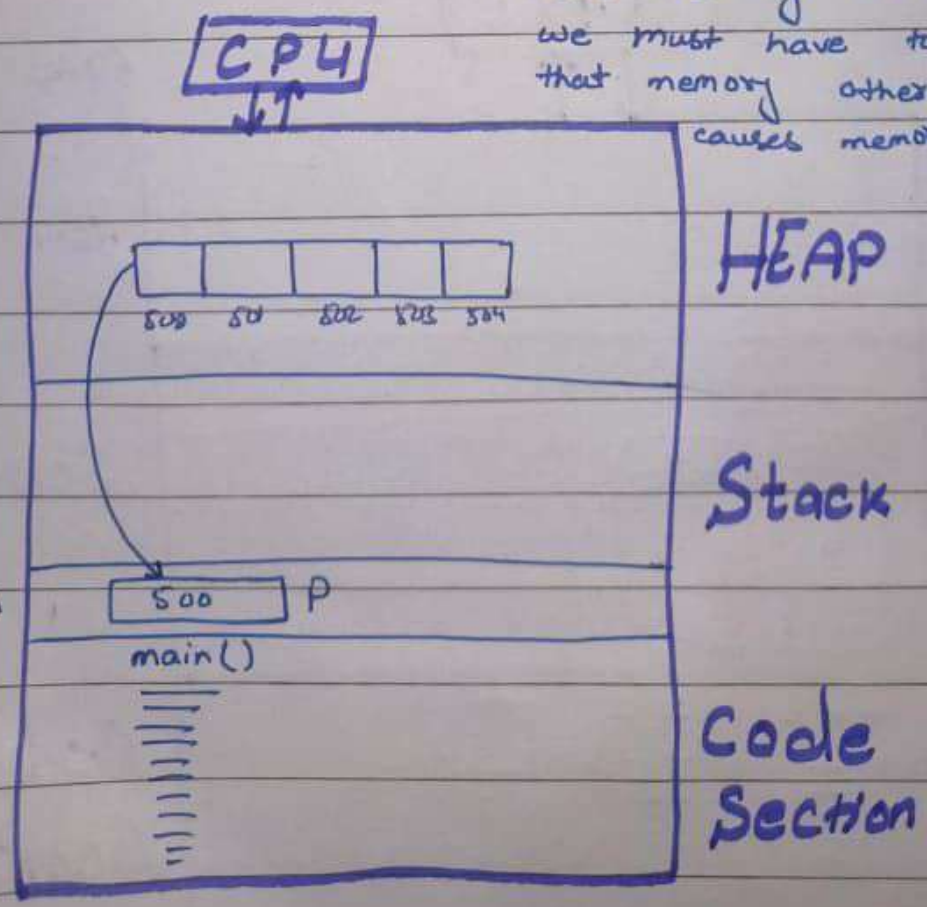
Dynamic Memory Allocation :-

Dynamic memory Allocation is always created in the heap Section of main memory.

Program cannot directly Access heap memory. So to Access heap memory we use pointer.

Program :-

```
void main () {
    int * ptr ;
    Ptr = new int [5]. In C++
    delete [] Ptr [ if we have executed the memory in heap we must have to delete that memory otherwise it causes memory loss ]
```